A.V.KAVERI,

192512188.

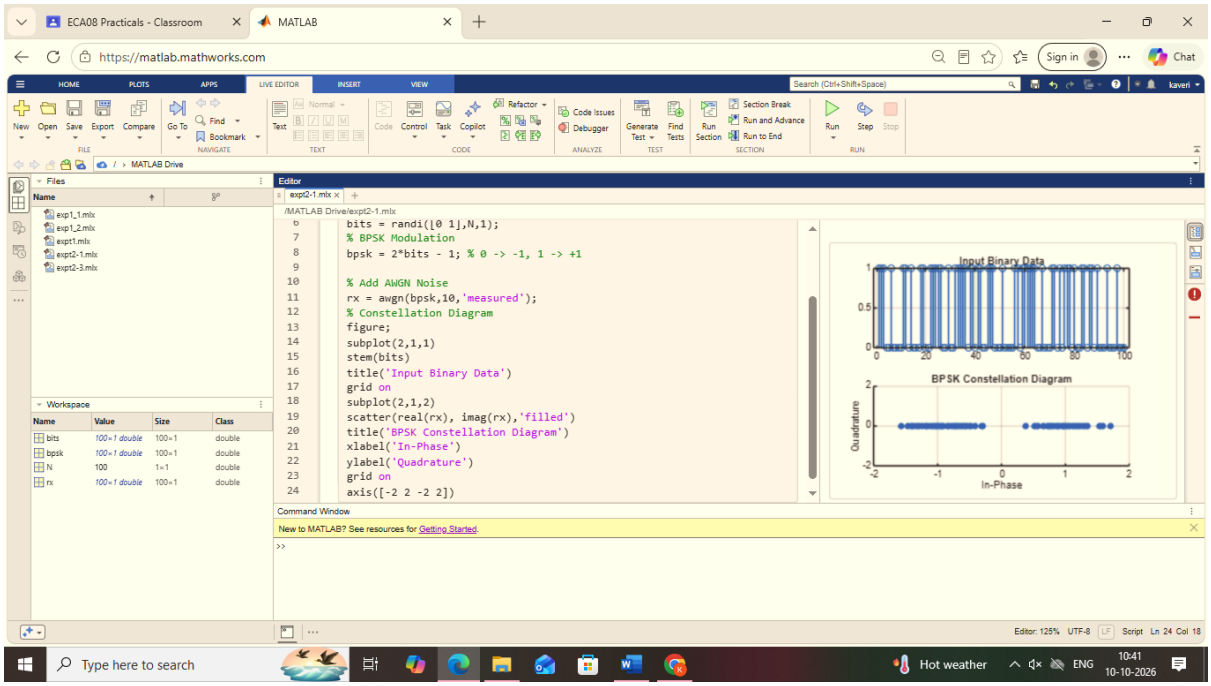
EXPERIMENT – 22

SIMULATION AND ANALYSIS OF BPSK MODULATION USING CONSTELLATION DIAGRAMS IN MATLAB

PROGRAM:

```
clc;
clear;
close all;
% Generate Random Bits
N = 100;
bits = randi([0 1],N,1);
% BPSK Modulation
bpsk = 2*bits - 1; % 0 -> -1, 1 -> +1
% Add AWGN Noise
rx = awgn(bpsk,10,'measured');
% Constellation Diagram
figure;
subplot(2,1,1)
stem(bits)
title('Input Binary Data')
grid on
subplot(2,1,2)
scatter(real(rx), imag(rx),'filled')
title('BPSK Constellation Diagram')
xlabel('In-Phase')
ylabel('Quadrature')
grid on
axis([-2 2 -2 2])
```

OUTPUT:


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MATLAB
https://matlab.mathworks.com
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Chat
HOME
PLOTS
APPS
LIVE EDITOR
INSERT
VIEW
MATLAB Drive
Files
exp1_1.mlx
exp1_2.mlx
expt1.mlx
expt2-1.mlx
expt2-3.mlx
Workspace
Name
Value
Size
Class
bits
bpsk
N
rx
Editor
expt2-1.mlx
/MATLAB Drive/expt2-1.mlx
bits = randi([0 1],N,1);
% BPSK Modulation
bpsk = 2*bits - 1; % 0 -> -1, 1 -> +1
% Add AWGN Noise
rx = awgn(bpsk,10,'measured');
% Constellation Diagram
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stem(bits)
title('Input Binary Data')
grid on
subplot(2,1,2)
scatter(real(rx), imag(rx),'filled')
title('BPSK Constellation Diagram')
xlabel('In-Phase')
ylabel('Quadrature')
grid on
axis([-2 2 -2 2])
Input Binary Data
BPSK Constellation Diagram
Quadrature
In-Phase
Command Window
New to MATLAB? See resources for Getting Started.
Type here to search
Hot weather
ENG
10:41
10-10-2026